

MEGA: MELODY-GUIDED AND HARMONY-ALIGNED VOCAL ACCOMPANIMENT GENERATION

Yuxuan Xia^{1,*}, Guobin Ma^{1,*}, Hongyu Lyu¹, Yuepeng Jiang¹,
Huakang Chen¹, Mengqi Wei¹, Liumeng Xue^{2,3}, Lei Xie^{1,**}

¹ Audio, Speech and Language Processing Group (ASLP@NPU),

School of Computer Science, Northwestern Polytechnical University, Xi'an, China

² School of Intelligence Science and Technology, Nanjing University, Suzhou, China

³ State Key Laboratory of Novel Software Technology, Nanjing University, Nanjing, China

ABSTRACT

Vocal-to-accompaniment generation (V2A) aims to generate instrumental accompaniment that musically matches a given vocal performance. Existing V2A methods typically take as input either vocal audio representations or simplified melody representations extracted from the same vocal. The former entangle melody with timbre and other acoustic factors, while the latter discard potentially useful vocal context. Moreover, explicit optimization for vocal-accompaniment compatibility remains limited. In this study, we propose MEGA, a **ME**lody-**G**uided and **H**armony-**A**ligned V2A framework built upon the pretrained ACE-Step 1.5. MEGA injects melody cues into the backbone's vocal-conditioned hidden states through a zero-gated cross-attention plugin, preserving rich vocal context while providing explicit melodic guidance. Factorized guidance independently adjusts text and melody contributions to balance text-driven arrangement with melodic adherence. We further employ harmony-aware direct preference optimization to improve both musical quality and vocal-accompaniment compatibility. Experimental results show that MEGA outperforms the compared V2A systems in overall musical quality and vocal-accompaniment compatibility. Audio demos are available at <https://aslp-lab.github.io/MEGA/>, and the code will be released as open source.

Index Terms— vocal-to-accompaniment generation, classifier-free guidance, direct preference optimization, music generation

1. INTRODUCTION

Creating an accompaniment around an existing vocal performance is a central part of music production, yet it demands close coordination of harmony, rhythm, phrasing, and style. Vocal-to-accompaniment generation (V2A) seeks to automate this process by synthesizing an instrumental backing conditioned on a given vocal recording. It simplifies music arrangement, enabling rapid song prototyping and personalized accompaniment. Recent neural V2A systems have shown promising results by conditioning generative models on vocal audio or derived melody representations [1, 2, 3, 4, 5]. Nevertheless, generating an accompaniment that is both musically compelling and closely matched to the input vocal remains challenging.

Existing V2A systems differ in how they expose the musical content of the vocal condition. ACE-Step 1.5 and LaDA-Band

operate directly on acoustic representations, preserving melodic, phonetic, and expressive cues but leaving the model to infer which of these entangled cues should govern the accompaniment [3, 4]. AnyAccomp instead compresses a vocal-derived chromagram into a quantized, largely timbre-invariant melody representation. This emphasizes melodic structure and mitigates separation artifacts, but discards part of the original performance context such as vocal timbre and fine-grained expressive cues [5]. A key challenge is therefore to provide clear melodic guidance while preserving the rich performance context of the input vocal. At the same time, cover versions of the same song often retain its core vocal melody but use different instruments and arrangements [6], highlighting an inherent one-to-many relationship between vocal melody and accompaniment. However, this relationship has received limited attention in existing V2A research.

Recent music generation models have adopted reinforcement learning (RL) and preference optimization to improve generation quality and controllability [3, 7, 8]. When these methods are applied to V2A, however, rewards or preferences derived solely from general-purpose music-quality scores may favor overall musical quality without explicitly accounting for vocal-accompaniment compatibility. Improving overall musical quality while ensuring compatibility with the input vocal therefore remains a challenge in V2A.

To address these challenges, we propose **MEGA**, a melody-guided V2A framework built upon a pretrained music foundation model. Our main contributions are as follows:

- We propose an **External Melody Modulation Module** that injects melody cues into the frozen backbone's vocal-conditioned hidden states through zero-gated cross-attention, making melodic structure explicit while preserving rich vocal acoustic context.
- We propose **Backbone Plugin Factorized Guidance** to independently adjust melody and backbone text guidance, enabling their contributions to be balanced during inference for improved song quality.
- To improve vocal-accompaniment compatibility alongside musical quality, we introduce **Harmony-Aware Preference Optimization**, which combines music-quality ranking with compatibility filtering to construct preference pairs for direct preference optimization (DPO).
- Objective and subjective evaluations demonstrate gains over the compared V2A systems in overall song quality and vocal-accompaniment compatibility, while ablations validate the benefits of the proposed components.

*These authors contributed equally.

** indicates the corresponding author.

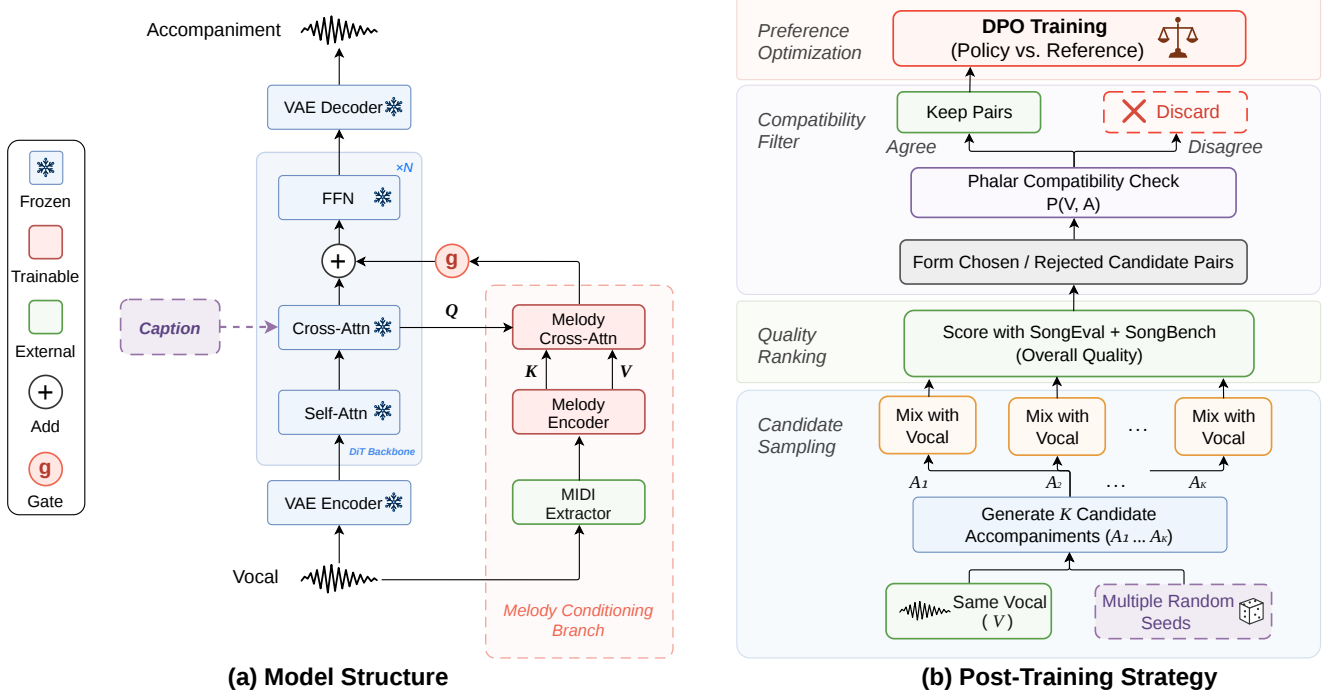


Fig. 1. Overview of MEGA: (a) vocal-conditioned accompaniment generation with melody and text guidance; (b) preference-pair construction by ranking candidate mixtures for song quality, then filtering the resulting pairs for vocal–accompaniment compatibility before DPO training.

2. METHOD

MEGA extends the pretrained ACE-Step 1.5 [3] with an explicit melody-conditioning branch, as shown in Fig. 1(a). The source vocal retains its original acoustic path through the frozen variational autoencoder (VAE) [9] and diffusion transformer (DiT) [10], while the caption is supplied through the backbone’s native conditioning interface. In parallel, we extract MIDI from the source vocal as explicit melody cues and encode it into melody states. These states modulate every DiT block after its native cross-attention and before its feed-forward network (FFN). The resulting latent is decoded by the frozen VAE decoder to produce the accompaniment.

2.1. External Melody Modulation Module

We transcribe the source vocal into MIDI using SOME [11] and expand the note IDs according to their durations into a frame-level sequence $\mathbf{m} \in \mathbb{N}^T$ at approximately 25 Hz, matching the audio latent rate. The melody encoder embeds $\mathbf{m}$ and applies a 1-D convolution with kernel size and stride equal to the DiT patch size, followed by two local convolutional layers to capture neighboring note context. Its output is interpolated or truncated to match the number of audio latent patches, yielding

$$\mathbf{Z} = E_M(\mathbf{m}) \in \mathbb{R}^{T_p \times d_M}, \quad (1)$$

where T_p is the number of latent patches and d_M is the melody embedding dimension. We subsequently inject $\mathbf{Z}$ into every DiT block after its native cross-attention. Let $\mathbf{H}_\ell$ denote the hidden states at this position in block ℓ . Using $\mathbf{H}_\ell$ as queries and $\mathbf{Z}$ as keys and values, we compute cross-attention [12]:

$$\begin{aligned} \mathbf{A}_\ell &= \text{CrossAttn}_\ell(\mathbf{H}_\ell, \mathbf{Z}, \mathbf{Z}), \\ \tilde{\mathbf{H}}_\ell &= \mathbf{H}_\ell + \tanh(g_\ell) \mathbf{A}_\ell. \end{aligned} \quad (2)$$

The learnable scalar gate g_ℓ is initialized to zero, making each modulation layer an identity mapping at initialization. Throughout training, the VAE, condition encoders, and DiT backbone remain frozen, and only E_M and the layer-wise melody modules are updated.

2.2. Backbone Plugin Factorized Guidance

The explicit plugin branch further allows melody guidance to be separated from the backbone’s native text guidance [13]. All predictions below share the same noisy latent $\mathbf{x}_t$ and timestep t , which are omitted for brevity. We define

$$\begin{aligned} \mathbf{v}_{CM} &= f(S, C, M), \quad \mathbf{v}_{C0} = f(S, C, 0), \\ \mathbf{v}_{\emptyset 0} &= f(S, \emptyset, 0). \end{aligned} \quad (3)$$

Here, S is the source-vocal acoustic context shared by all branches; C is the backbone’s text condition; and M is the MIDI condition. The symbol 0 denotes an empty melody input, whereas $\emptyset$ replaces the condition C with its null embedding.

Our Backbone Plugin Factorized Guidance (BPGF) predicts the final velocity as

$$\hat{\mathbf{v}} = \text{APG}_\gamma(\mathbf{v}_{C0}, \mathbf{v}_{\emptyset 0}) + \lambda_M(\mathbf{v}_{CM} - \mathbf{v}_{C0}), \quad (4)$$

where APG_γ applies adaptive projected guidance [14] to the condition C with scale γ . The difference $\mathbf{v}_{CM} - \mathbf{v}_{C0}$ isolates the incremental velocity introduced by the melody plugin while holding S and C fixed, and λ_M controls its strength. Thus, γ controls text adherence and λ_M independently controls melody following.

2.3. Harmony-Aware Preference Optimization

Figure 1(b) summarizes how we construct our preference data. For each input vocal, we generate multiple candidate accompaniments using different random seeds and mix each with the original vocal. We then evaluate the mixtures using SongEval and SongBench [15,

16]. To account for their different score ranges, we halve the mean SongBench score and average it with the mean SongEval score to obtain an overall quality score. We form preference pairs only when the chosen candidate outperforms the rejected candidate by minimum margins in overall quality, SongEval, and normalized SongBench.

These evaluators measure the aggregate quality of a song mixture, but do not directly determine whether one accompaniment is better matched to the given vocal than another. To complement these general-purpose scores, we introduce a dedicated compatibility filter based on PHALAR [17], a contrastive model originally developed for stem-to-mix retrieval. PHALAR encodes harmonic features into complex-valued representations whose phase preserves temporal information, and computes pairwise scores using a learned bilinear similarity function. Since its original training setup is not tailored to vocal-accompaniment matching, we train it on paired vocal and accompaniment tracks, using time-aligned segments from the same song as positive pairs and other accompaniment segments within the batch as negatives. This training encourages higher scores for aligned pairs, allowing us to assess each candidate’s compatibility with the input vocal without requiring a reference accompaniment.

We then use these compatibility scores to further filter the quality-based preference pairs. A pair is retained only if PHALAR also favors the chosen candidate, its compatibility score meets a minimum threshold, and its advantage over the rejected candidate meets a minimum margin. Finally, we freeze the reference plugin and update only the policy plugin with DPO [18, 19], enlarging the policy-to-reference preference margin between the chosen and rejected accompaniments.

3. EXPERIMENTS

3.1. Experimental Setup

Datasets. For supervised training, we use an in-house dataset of 6,805 vocal-accompaniment stem pairs sourced from licensed, non-AI-generated songs. For each pair, the accompaniment is processed by the `understandmusic` pipeline of ACE-Step 1.5 to obtain the caption. SOME [11] transcribes the corresponding vocal into the MIDI condition. For preference optimization, we reuse all training inputs and generate eight candidate accompaniments per input using different random seeds. We mix each candidate with the original vocal and evaluate the resulting mixture using the combined quality score defined in Sec. 2.3. For each input, we select the highest- and lowest-scoring candidates as the chosen and rejected samples, respectively. We retain a pair only if the chosen sample exceeds the rejected sample by at least 0.50 in the combined quality score and by at least 0.10 in both SongEval and normalized SongBench. To account for vocal-accompaniment compatibility, we further require the chosen sample’s PHALAR compatibility score to be at least 0.8 and to exceed the rejected sample’s score by at least 0.2. The final DPO training set contains 706 preference pairs.

Evaluation is conducted on two multi-genre test sets, each containing 500 songs: an in-house real-world set and a subset of Suno70K [6]. For both sets, we use the Band RoFormer separator [20] provided by YingMusic-SVC [21] to obtain vocal and accompaniment stems. For the real-world set, captions are obtained in the same way as for the training data. For Suno70K, we directly use the captions provided with the dataset as input. For both sets, MIDI conditions are extracted from the separated vocals using SOME.

Implementation Details. The melody encoder uses a hidden size of $d_M = 256$. Each layer-wise cross-attention module operates in

a 1,024-dimensional interaction space and uses eight attention heads with a head dimension of 128. Overall, the melody plugin contains 114.2M trainable parameters, while the ACE-Step backbone remains frozen.

We first train the melody plugin with supervised learning using the AdamW optimizer, with a per-GPU batch size of 8 and a learning rate of 5×10^{-5} . Starting from the supervised checkpoint at 15,000 steps, we further optimize the plugin with DPO. This stage uses the same optimizer, with a per-GPU batch size of 16 and a learning rate of 1×10^{-5} . We evaluate the checkpoint after 500 DPO steps. Both training stages are conducted on eight NVIDIA L40 GPUs.

Evaluation Metrics. We mix each generated accompaniment with its input vocal and use SongEval [15], SongBench [16], and AudioBox Aesthetics [22] to assess the overall quality of the resulting song. Beyond overall song quality, we use Accompaniment Prompt Adherence (APA) [23] to evaluate the compatibility between the generated accompaniment and the input vocal. We also use Fréchet Audio Distance (FAD) [24] to measure distributional similarity between generated and reference accompaniments. Higher values are better for all objective metrics except FAD.

For subjective evaluation, we design three complementary criteria to assess the perceptual quality of the generated accompaniment, considering both its intrinsic musical qualities and its role in supporting the input vocal. Audio Quality (AQ) assesses the clarity and naturalness of the accompaniment and the absence of audible artifacts. Arrangement Musicality (AM) evaluates the musical coherence of the accompaniment itself, including its rhythm, harmony, instrumentation, and structure. Vocal-Accompaniment Compatibility (VAC) assesses how well the accompaniment supports the vocal in harmony, rhythm, and phrasing. All three dimensions are rated on a five-point scale (1: very poor; 5: excellent).

Compared Systems. To validate the performance of MEGA, we compare it against several strong and representative V2A systems, including ACE-Step 1.5 [3], LaDA-Band [4], and AnyAccomp [5]. For AnyAccomp, which operates on short audio segments and supports only vocal-derived conditioning, we perform inference in 30-second chunks and concatenate the outputs for full-song evaluation. The ground-truth mixtures are also evaluated for reference.

3.2. Objective Evaluation

Table 1 compares MEGA with three representative V2A systems on Suno70K and Real-world. Among the evaluated systems, MEGA achieves the highest SongEval, SongBench, and APA scores on both datasets. It also obtains competitive FAD and AudioBox Aesthetics [22] results. The strong SongEval and SongBench results indicate that MEGA generates accompaniments that form coherent and musically convincing songs when combined with the input vocals, while its leading APA scores suggest improved vocal-accompaniment compatibility.

Compared with its pretrained ACE-Step 1.5 backbone, MEGA consistently improves across all objective metrics on both datasets. These results support the effectiveness of incorporating explicit melodic guidance and harmony-aware preference optimization into the frozen backbone. Although MEGA does not achieve the best AudioBox Aesthetics scores among all systems, it consistently improves over ACE-Step 1.5 and remains close to the strongest competitors. Since the ACE-Step backbone is kept frozen throughout training, the overall aesthetic quality may still be partly constrained by the generative capability of the pretrained backbone.

3.3. Subjective Evaluation

We conducted a blind listening test on 50 songs selected from Suno70K and Real-world, comparing MEGA, ACE-Step 1.5, LaDA-

Table 1. Objective results on Suno70K and Real-world and pooled subjective scores, reported as mean opinion score (MOS) $\pm$ 95% confidence interval (CI) half-width. SE, SB, and AB denote SongEval, SongBench, and AudioBox Aesthetics. Best and second-best generated scores are bold and underlined.

Method	Suno70K					Real-world					Subjective (MOS $\pm$ 95% CI)		
	SE $\uparrow$	SB $\uparrow$	AB $\uparrow$	APA $\uparrow$	FAD $\downarrow$	SE $\uparrow$	SB $\uparrow$	AB $\uparrow$	APA $\uparrow$	FAD $\downarrow$	AQ $\uparrow$	AM $\uparrow$	VAC $\uparrow$
Ground Truth	4.34	6.33	7.40	1.00	0.00	3.97	6.25	7.25	1.00	0.00	—	—	—
ACE-Step 1.5	3.34	5.24	6.66	<u>0.26</u>	0.61	3.07	5.27	6.91	<u>0.51</u>	0.54	3.45 ± 0.16	3.40 ± 0.17	3.55 ± 0.16
LaDA-Band	<u>3.81</u>	<u>5.66</u>	7.19	0.10	0.35	<u>3.59</u>	<u>5.72</u>	<u>7.19</u>	0.33	0.29	<u>3.89 ± 0.13</u>	<u>3.82 ± 0.14</u>	<u>3.78 ± 0.15</u>
AnyAccomp	3.40	5.54	7.28	0.00	0.70	3.22	5.62	7.20	0.22	0.70	3.32 ± 0.17	3.45 ± 0.16	3.36 ± 0.18
MEGA (Ours)	3.98	6.17	<u>7.20</u>	0.52	<u>0.54</u>	3.66	5.96	7.17	0.75	<u>0.42</u>	4.18 ± 0.12	4.25 ± 0.11	4.27 ± 0.12

Band, and AnyAccomp. Twenty listeners, including 13 with a musical background and seven without, rated the samples on AQ, AM, and VAC.

As shown in Table 1, MEGA receives the highest MOS across all three dimensions. The gains in AQ and AM suggest improved perceptual fidelity and musical coherence in the generated accompaniments. Higher VAC scores further indicate that these accompaniments better support the input vocals than other compared systems in harmony, rhythm, and phrasing. These findings are consistent with the objective evaluation.

Table 2. Ablation study of MEGA on Suno70K and Real-world.

Variant	Suno70K			Real-world		
	SE $\uparrow$	SB $\uparrow$	APA $\uparrow$	SE $\uparrow$	SB $\uparrow$	APA $\uparrow$
ACE-Step Base	3.34	5.24	0.26	3.07	5.27	0.51
ACE-Step + LoRA	3.52	5.77	0.29	3.11	5.47	0.51
ACE-Step + Melody Plugin						
Flat Combined CFG	3.70	5.98	0.30	3.31	5.70	0.56
BPFG ($\lambda_M = 1$)	3.61	5.88	0.29	3.28	5.65	0.58
BPFG ($\lambda_M = 3$)	3.80	6.06	0.34	3.47	5.79	0.56
+ DPO w/o PHALAR	3.95	6.16	0.41	3.57	5.87	0.62
+ DPO w/ PHALAR (MEGA)	3.98	6.17	0.52	3.66	5.96	0.75

3.4. Ablation Study

Melody Module. We compare ACE-Step Base, ACE-Step with low-rank adaptation (LoRA) [25], and ACE-Step + Melody Plugin using their corresponding entries in Table 2. For the melody plugin, we use the BPFG ($\lambda_M = 1$) row before DPO. LoRA and the melody plugin are trained on the same vocal-accompaniment data. Both adaptations improve SongEval and SongBench over the base model. Compared with LoRA, the melody plugin achieves higher SongEval and SongBench scores on both datasets and higher APA on Real-world, with comparable APA on Suno70K. These results suggest that the proposed melody-conditioning architecture provides additional benefits beyond fine-tuning on the same data alone.

Backbone Plugin Factorized Guidance. We compare Flat Combined classifier-free guidance (CFG) and BPFG using the same melody plugin before DPO. BPFG separately weights the melody residual as defined in Eq. 4, whereas Flat Combined applies APG to the joint text and melody conditions:

$$\hat{\mathbf{v}}_{\text{flat}} = \text{APG}_{\gamma}(\mathbf{v}_{CM}, \mathbf{v}_{\emptyset 0}). \quad (5)$$

The melody plugin introduces explicit cues into a backbone that already receives vocal acoustic context. Learning to incorporate

these cues does not, by itself, establish how strongly their contribution should be weighted during sampling. BPFG provides a separate control over this learned contribution while retaining the backbone’s native text-guidance setting.

The three guidance settings in Table 2 illustrate the importance of this adjustment. BPFG with $\lambda_M = 1$ yields lower SongEval and SongBench scores than Flat Combined CFG on both datasets, whereas $\lambda_M = 3$ achieves the highest scores among these three settings. This reversal suggests that factorization needs to be accompanied by appropriate weighting of the melody residual to improve generation quality. In the evaluated setting, giving the learned melody contribution greater emphasis improves the overall quality of the resulting vocal-accompaniment mixtures without additional training. These results support the complementary roles of the melody plugin and BPFG: the former makes explicit melodic information available, while the latter adjusts how strongly that information influences generation.

Harmony-Aware Preference Optimization. We compare the model before DPO with the two DPO variants in Table 2, all using BPFG ($\lambda_M = 3$). Using quality-based preferences alone, DPO improves SongEval and SongBench on both datasets, suggesting that preference optimization further enhances song quality beyond supervised plugin training.

Adding PHALAR filtering further improves APA on both datasets, with gains exceeding those obtained from quality-based DPO alone. SongEval and SongBench also improve, indicating that the additional emphasis on vocal compatibility extends the quality benefits of DPO. These results support the complementary role of PHALAR in preference construction: while SongEval and SongBench favor mixtures with higher overall quality, PHALAR additionally requires the preferred accompaniment to better match the input vocal. Incorporating this criterion helps direct preference optimization toward vocal fit alongside overall song quality.

4. CONCLUSION

In this study, we presented MEGA, a melody-guided V2A framework built upon the pretrained ACE-Step 1.5. MEGA complements vocal acoustic context with explicit melody cues and enables independent control of text and melody guidance during inference. Harmony-aware DPO further improves musical quality and vocal-accompaniment compatibility. Experimental results show that MEGA achieves the best SongEval, SongBench, and APA scores on both test sets and receives the highest listener ratings across all three subjective dimensions. Together, these results demonstrate that MEGA outperforms the compared V2A systems in both musical quality and vocal-accompaniment compatibility.

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6. COMPLIANCE WITH ETHICAL STANDARDS

The listening test involved volunteers who were informed of the research purpose and agreed to participate. Ratings were collected anonymously, and no personally identifiable information was recorded. The test was conducted in accordance with the Declaration of Helsinki (1975, revised 2000).

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